

Defective CAPSL function causes impaired retinal angiogenesis through the MYC axis and is associated with familial exudative vitreoretinopathy

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Abstract

Familial exudative vitreoretinopathy (FEVR) is a severe genetic disorder characterized by incomplete vascularization of the peripheral retina and associated symptoms that can lead to vision loss. However, the underlying genetic causes of approximately 50% of FEVR cases remain unknown. Here, we report two heterozygous variants, c.88C>T (p.Arg30Ter) and c.247C>T (p.Leu83Phe), in calcyphosine like (*CAPSL*), from four patients in two unrelated FEVR-affected families. Both variants exhibited compromised *CAPSL* protein expression. Vascular endothelial cell-specific inactivation of *Capsl* in postnatal mice resulted in defective sprouting, delayed radial/vertical vascular progression, compromised endothelial proliferation, and impaired cell migration, recapitulating the human FEVR phenotypes. *CAPSL*-depleted human retinal microvascular endothelial cells (HRECs) exhibited impaired tube formation, decreased cell proliferation, disrupted cell polarity establishment and filopodia/lamellipodia formation, as well as disrupted collective cell migration *in vitro*. Transcriptomic and proteomic profiling of *CAPSL*-depleted HRECs revealed that *CAPSL* abolition inhibited the MYC signaling axis, in which the expression of core MYC targeted genes were profoundly decreased. Furthermore, a combined analysis of *CAPSL*-depleted HRECs and *c-MYC*-depleted human umbilical vein endothelial cells (HUVECs) uncovered similar transcription patterns. Collectively, this study reports a novel FEVR-associated candidate gene, *CAPSL*, which provides invaluable information for genetic counseling and prenatal diagnosis of FEVR. This study also reveals that compromised *CAPSL* function causes FEVR through MYC axis, shedding light on the potential involvement of MYC signaling in the pathogenesis of FEVR.

eLife assessment

This study explores the role of calcyphosine-like (CAPSL) in Familial Exudative Vitreoretinopathy (FEVR) via the MYC pathway, offering **valuable** insights into disease mechanisms that are supported by a **solid**, multi-pronged approach. The overall significance of the study might, however, be limited due to weak support from human genetic studies.

Introduction

A well-organized vascular system is crucial for the proper development of most tissue and organ morphogenesis^[1,2]. Consequently, angiogenic defects have been associated with several congenital human diseases^[3,4]. Familial exudative vitreoretinopathy (FEVR) is a heritable vitreoretinopathy characterized by defective retinal angiogenesis and various complications, such as extensive neovascularization, exudation, retinal folds and detachments, and vision loss⁵. Based on an ocular screening of over sixty thousand Chinese newborns, the estimated prevalence of FEVR is around 0.46%⁶. Due to the significant genetic heterogeneity, FEVR displays all Mendelian forms of inheritance: autosomal dominant (AD), autosomal recessive (AR), or X-linked recessive (XR)^[7-9]. To date, mutations in 17 genes and 1 locus have been identified to cause FEVR, including Norrin (*NDP*)¹⁰, frizzled 4 (*FZD4*)¹¹, low-density lipoprotein receptor related protein 5 (*LRP5*)^[12,13], low-density lipoprotein receptor-related protein 6 (*LRP6*)¹⁴, tetraspanin-12 (*TSPAN12*)^[15,16], α-catenin (*CTNNA1*)¹⁷, β-catenin (*CTNNB1*)^[18-20], p120-catenin (*CTNND1*)²¹, zinc finger protein 408 (*ZNF408*)²², kinesin family member 11 (*KIF11*)²³, atonal homolog 7 (*ATOH7*)²⁴, exudative vitreoretinopathy 3 (*EVR3*)²⁵, integrin-linked kinase (*ILK*)²⁶, jagged canonical Notch ligand 1 (*JAG1*)²⁷, discs large MAGUK scaffold protein 1 (*DLG1*)²⁸, transforming growth factor beta receptor 2 (*TGFBR2*)²⁹, sorting nexin 31 (*SNX31*)³⁰, and ER membrane protein complex subunit 1 (*EMC1*)³¹. Nevertheless, these mutations can explain only approximately 50% of FEVR cases^[32,33].

Calcyphosine like (*CAPSL*) is a protein-coding gene whose function remains elusive. In 2019, Julia Schreml pointed out that *CAPSL* is a promising candidate gene for multiple symmetric lipomatosis (MSL), a rare adipose tissue disorder of largely unknown etiology³⁴. Nevertheless, the molecular mechanisms by which *CAPSL* regulates cell processes and signaling cascades have yet to be elucidated.

MYC gene family is one of the most widely investigated cancer-causing genes, being implicated in the formation, maintenance, and progression of several different cancer types, such as breast cancer, lymphoma, and prostatic cancer^[35-38]. *c-MYC* is a master regulator in *MYC* signaling, which participates in cell metabolism, proliferation, differentiation, and migration³⁹. The important role of *c-Myc* in vascularization and angiogenesis during tumor development has already been reported in mouse models^[40,41]. Vascular endothelial cell (EC)-specific knockout of *c-Myc* in mice leads to impaired vascular expansion, thinned and poorly branched vascular, and reduced endothelial proliferation⁴². Conversely, EC-specific *c-Myc* over-expression caused sustained vascular outgrowth, increased endothelial cell proliferation, and vessel density⁴². These studies indicate that *c-MYC* is a regulating factor in coordinating endothelial cell behavior and vessel development.

In the current study, from a large cohort of FEVR patients, we identified a missense mutation and a stop-gain mutation in the *CAPSL* gene from four FEVR patients by whole exome sequencing (WES) analysis. Furthermore, the endothelial cell (EC)-specific *Capsl* knockout mouse model exhibited

FEVR-like retinal vascular defects, emphasizing the indispensable role of *Capsl* for the retinal vascular architecture. Knockdown of *CAPSL* led to compromised proliferation of stalk ECs and retarded migration of tip ECs by regulating the MYC signaling axis. Collectively, this study not only identifies *CAPSL* as a candidate gene for FEVR, but also presents that a compromised MYC signaling axis might constitute a previously unrecognized factor contributing to FEVR.

Results

WES analysis revealed two *CAPSL* variants in FEVR patients

To evaluate the causative variants that account for FEVR, we applied WES on genomic DNA samples from FEVR families. Sanger sequencing was further applied to validate the variants and genotype-phenotype co-segregation analysis. Thus, two novel candidate variants in the *CAPSL* gene (NM_144647) were identified in patients with FEVR, and the variants were predicted to be pathogenic by Mutation taster, Polyphen-2, and PROVEAN (Table S1). In family 3036, a heterozygous c.88C>T (p.R30X) variant was identified in an infant (age 0-5 years old) (**Figure 1A**). His father was a heterozygous carrier and manifested vascular defects in the retina, including peripheral neovascularization area and leakage by fundus fluorescein angiography, whereas the wild-type mother showed normal vision (**Figures 1B and D**). The affected amino acid is highly conserved among different species (**Figure 1C**). The proband (II:1) in the 3104 family has been diagnosed with FEVR and identified with a heterozygous variant c.247C>T (p.L83F) (**Figure 1A**). His father has also been identified as a carrier of this heterozygous variant, manifested with FEVR symptoms, according to the medical records.

To determine the pathogenesis of *CAPSL* variants for FEVR, we first investigated the effect of variants on the expression of *CAPSL*. Wild-type and mutant coding sequences of *CAPSL* were introduced to the GFP-tagged pcDNA3.1 vector, and an IRES2 box was inserted to prevent the fusion expression of *CAPSL* and GFP. Immunoblot analysis of cells transfected with plasmids revealed that the *CAPSL*-R30X protein was undetectable, and the L83F variant resulted in a considerably reduced protein level (**Fig.1E** and F). Conversely, neither variant influenced the mRNA levels of *CAPSL* (**Fig.1G**). Thus, these variants may cause FEVR by impairing *CAPSL* expression.

Depletion of *Capsl* in vascular endothelial cells caused FEVR-liked phenotypes

We generated an inducible endothelial (EC)-specific *Capsl*-knockout mouse by breeding mice carrying a loxp-flanked allele of *Capsl* with tamoxifen-inducible *Pdgfrb-iCreER*⁴³ transgenic animals to determine whether depletion of *Capsl* in mouse endothelial cells could cause FEVR-like phenotypes (Fig.S1A). *Capsl*^{loxP/loxP}, *Pdgfrb-CreER* mice (hereafter named *Capsl*^{iECKO/iECKO}) and their control littermates (*Capsl*^{loxP/loxP} and *Pdgfrb-CreER*, hereafter named Ctrl) were induced by consecutive intraperitoneal injection of 50 µg tamoxifen per pup at postnatal day 1 (P1) to P3 before sacrifice (Fig.S1B and C). *Capsl*^{iECKO/iECKO} mice showed indistinguishable overall appearance compared to their Ctrl littermates. The efficiency of EC-specific deletion of *Capsl* was confirmed by real-time quantitative PCR (RT-qPCR) and western blot analysis of P35 mouse lung tissues (Fig.S1D). The results showed a pronounced reduction in *CAPSL* expression, at both mRNA and protein levels, in *Capsl*^{iECKO/iECKO} mice compared to Ctrl littermates.

To assess the role of *CAPSL* in the growth and patterning of retinal vasculature, we employed flat-mount analysis to visualize the retinal vessels at different postnatal ages. As expected, the radial vascular growth, as well as vessel density and vascular branching, are dramatically reduced in P5 *Capsl*^{iECKO/iECKO} retina relative to Ctrl mice (**Fig.2A** and D-F). The number of retinal vascular tip

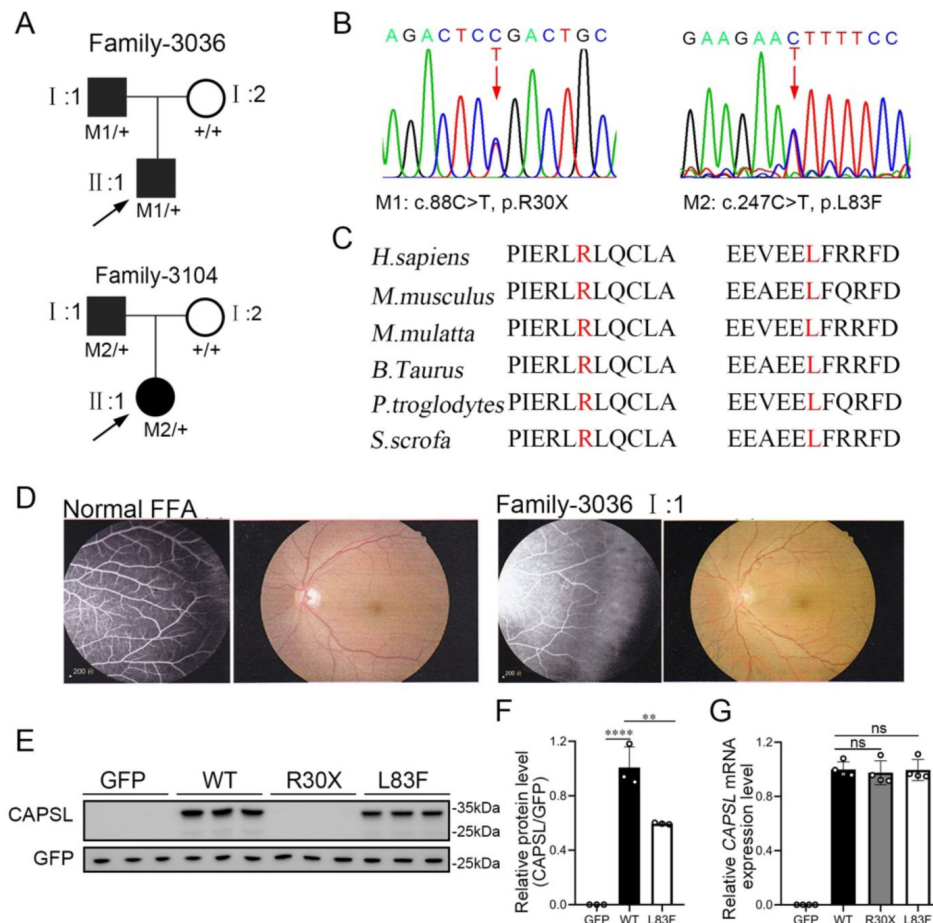


Figure 1.

CAPSL point mutations in two families with familial exudative vitreoretinopathy (FEVR).

(A-B) FEVR pedigrees (patients are denoted with black symbols) and Sanger sequencing of two heterozygous mutations identified in two families. Black arrows indicate the proband of each family and red arrows indicate the changed nucleotides. (C) Alignment of amino sequences surrounding the CAPSL variants in different species and all mutated sites are highly conserved. Altered amino acid residues are highlighted in red. (D) Fundus fluorescein angiography (FFA) (left panel) and fundus photography (right panel) of a normal individual and FEVR-affected patient (I:1) in family 3036. (E-G) Western blot (E-F) and RT-qPCR analysis (G) of CAPSL expression of WT and mutant plasmids. An empty vector with GFP tag was used as a negative control. GFP was used as an internal reference. Error bars indicate the SD.

P < 0.01, **P < 0.0001, ns: no significance, by Student's t test (n = 3).

cells, which leads EC migration toward high VEGFA by extending filopodia at the angiogenic front⁴⁴, was considerably decreased and sparsely distributed in *Capsl*^{iECKO/iECKO} retinæ compared to that of control littermates (**Fig.2B** [C](#) and [G-H](#)).

The onset of mouse retinal vascular development occurs postnatally at P0, coinciding with the regression of the hyaloid vasculature^[45,46]. Reproducing the hyaloid phenotypes observed in the FEVR-associated mouse model, the hyaloid vasculature in *Capsl*^{iECKO/iECKO} at P10 displayed delayed regression in comparison to the Ctrl littermates (**Fig.3A** [C](#)). Following the expansion of the superficial vessel plexus at around P8, retinal vessels extend perpendicularly into the deep retina, forming secondary and deep third capillary layers⁴⁷. To investigate whether the depletion of *Capsl* hinders deep capillary vascular development, we conducted immunostaining on P10 retinal sections. The results revealed a delayed development of deep retinal vessels in *Capsl*^{iECKO/iECKO} mouse retinæ compared to that of Ctrl (**Fig.3B** [C](#)). In addition, immunostaining of P14 flat-mount retina confirmed an incomplete architecture of the deep vascular layer in *Capsl*^{iECKO/iECKO} mouse (**Fig.3C** [C](#)).

The development of retinal vasculature is a complex process encompassing not only the coordinated behavior of endothelial cells but also the intricate interactions among different cell types and cytokines within the retina⁴⁸. Astrocytes form a mesh as a template prior to vascular growth and guide angiogenic expansion by secreting gradient distributed VEGFA^[49-51]. Abnormal activation of astrocytes and subsequent inflammatory responses have been observed in multiple FEVR mouse models^[15,17,47,52]. To explore whether the gradient and expression level of VEGFA were affected by the depletion of *Capsl*, we conducted immunostaining of VEGFA on the retina. However, the results showed no alteration of VEGFA (**Fig.S2A**). Meanwhile, the activation state and the pattern of the vascular template for EC outgrowth formed by astrocytes remained normal, indicated by GFAP (**Fig.S2B**). Pericyte recruitment is necessary for vessel stability and maturation⁵³. The immunostaining of Desmin or NG2 staining exhibited no difference between Ctrl and *Capsl*^{iECKO/iECKO} mice (**Fig.S2C**). To assess the integrity of the retinal vascular barrier, we performed the intraperitoneal injection of biocytin-TMR (869 Da), a small-molecular-weight fluorescent tracer⁵⁴. The results showed no notable leakage in the *Capsl*^{iECKO/iECKO} retina, suggesting the intact blood barrier despite the depletion of ECs-*Capsl* (**Fig.S2D**). The extracellular matrix (ECM) coordinates the behavior of ECs by regulating cell-cell communication processes⁵⁵. The deficiency and abnormal accumulation of ECM can both disrupt the adhesion, proliferation, and migration of the endothelial cells, leading to defects in blood vessel architecture^[56-58]. We thus analyzed the extracellular matrix accumulation using fibronectin 1, and the result showed normal deposition of ECM (**Fig.S2E**). In conclusion, endothelial cell-specific knockout of *Capsl* may mainly lead to defects in retinal vascular ECs rather than other vascular cells.

CAPSL controls vascular proliferation and migration

Considering the compromised vascular progression and decreased vessel density in *Capsl*^{iECKO/iECKO} mouse retinæ, we first performed the EdU assay on retina flat-mounts co-stained with ETS transcription factor ERG (endothelial cell nuclear marker) to assess the proliferation of ECs. The results showed a strongly reduced proliferation of *Capsl*-defective ECs compared to that of Ctrl (**Fig.4A** [C](#)). As previously reported, the regressed retinal vessels exhibited sleeve-like positivity for basement matrix component Collagen IV and negativity for Isolectin IB4⁵⁹. To compare the vascular regression between *Capsl*^{iECKO/iECKO} and Ctrl retinæ, we conducted co-staining of the Collagen IV and IB4. The results revealed that depletion of *Capsl* leads to an increased vascular regression, evidenced by a significant increase in collagen IV basal membrane sleeves lacking IB4 both in the capillary plexus and angiogenic front (**Fig.4B** [C](#)).

Additionally, considering the pivotal role of tip cells in the direction of vascular development, we then asked whether depletion of *Capsl* impairs the polarity of tip cells by analyzing the morphology of cell nuclei at the angiogenic front. This analysis was based on the observation that

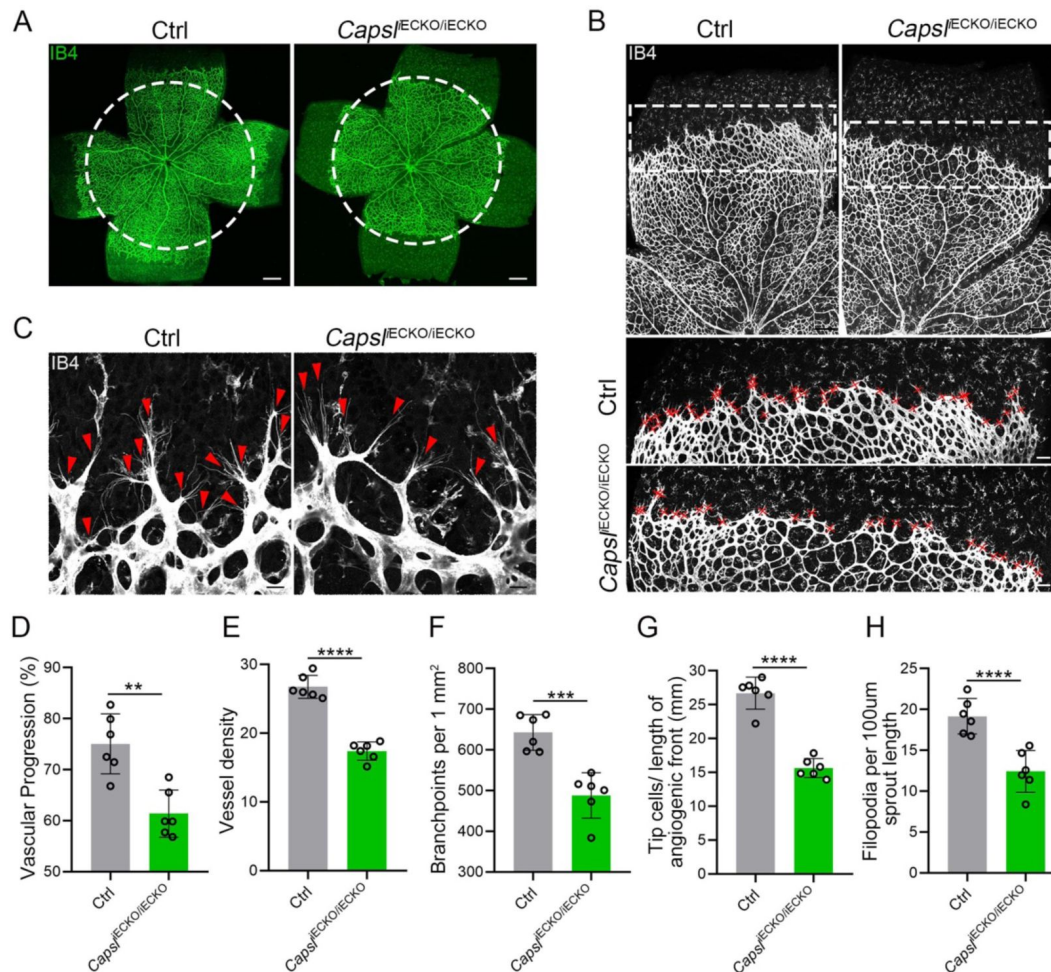


Figure 2.

EC-specific inactivation of *Capsl* impairs retina angiogenesis.

(A) Flat-mounted retinas obtained from P5 Ctrl and littermate *Capsl^{IECKO/IECKO}* mice were stained with Isolectin-B4 (IB4) to visualize blood vessel. Dashed circle mark the edge of the developing retina vessel in *Capsl^{IECKO/IECKO}* mice. Scale bar: 250 μ m. (B) Low magnification images (top panels) and high magnification images (bottom panels) in boxed areas of IB4-stained angiogenic front of Ctrl and *Capsl^{IECKO/IECKO}* mice, respectively. Red cross mark tip cells at the angiogenic growth front. Scale bar: 100 μ m (top panels), 25 μ m (bottom panels). (C) High magnification images of filopodia-extending cells at the edge of retinal angiogenic growth front from Ctrl and *Capsl^{IECKO/IECKO}* mice. Red arrowheads indicate the sprouts at the angiogenic growth front. (D-H) Quantification of retinal vascular development parameters, including vascular progression, vessel density, branchpoints, number of tip cells, and number of filopodia. Error bars indicate the SD. **P < 0.01, ***P < 0.001, ****P < 0.0001, by Student's t test (n = 6).

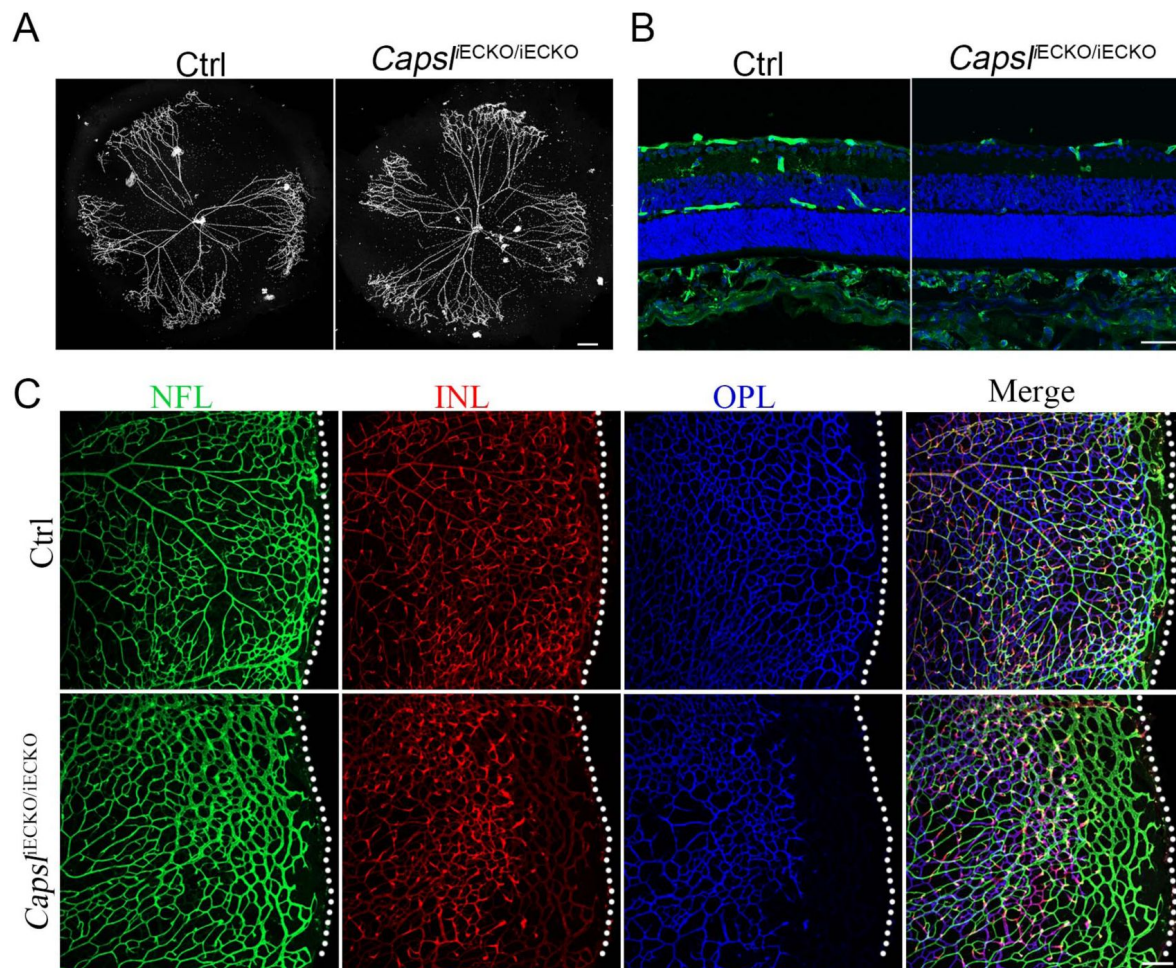


Figure 3.

Loss of *Capsl* results in delayed hyaloid regression and deep retinal blood vessel growth.

(A) Hyaloid vessels stained with DAPI in the eyes of Ctrl and *Capsl*^{IECKO/IECKO} mice at P10. Scale bar: 250 μ m. (B) Retina sections from P10 Ctrl and *Capsl*^{IECKO/IECKO} mice were costained with IB4 (green) and DAPI (blue). Scale bar: 100 μ m. (C) Flat-mounted retinas stained with IB4 at P14 Ctrl and *Capsl*^{IECKO/IECKO} mice. Optical sections of z-stacked confocal images were divided to represent the nerve fiber layer (NFL), inner plexiform layer (INL), and outer plexiform layer (ONL). Dashed lines mark the edge of the developing retina. Scale bar: 100 μ m

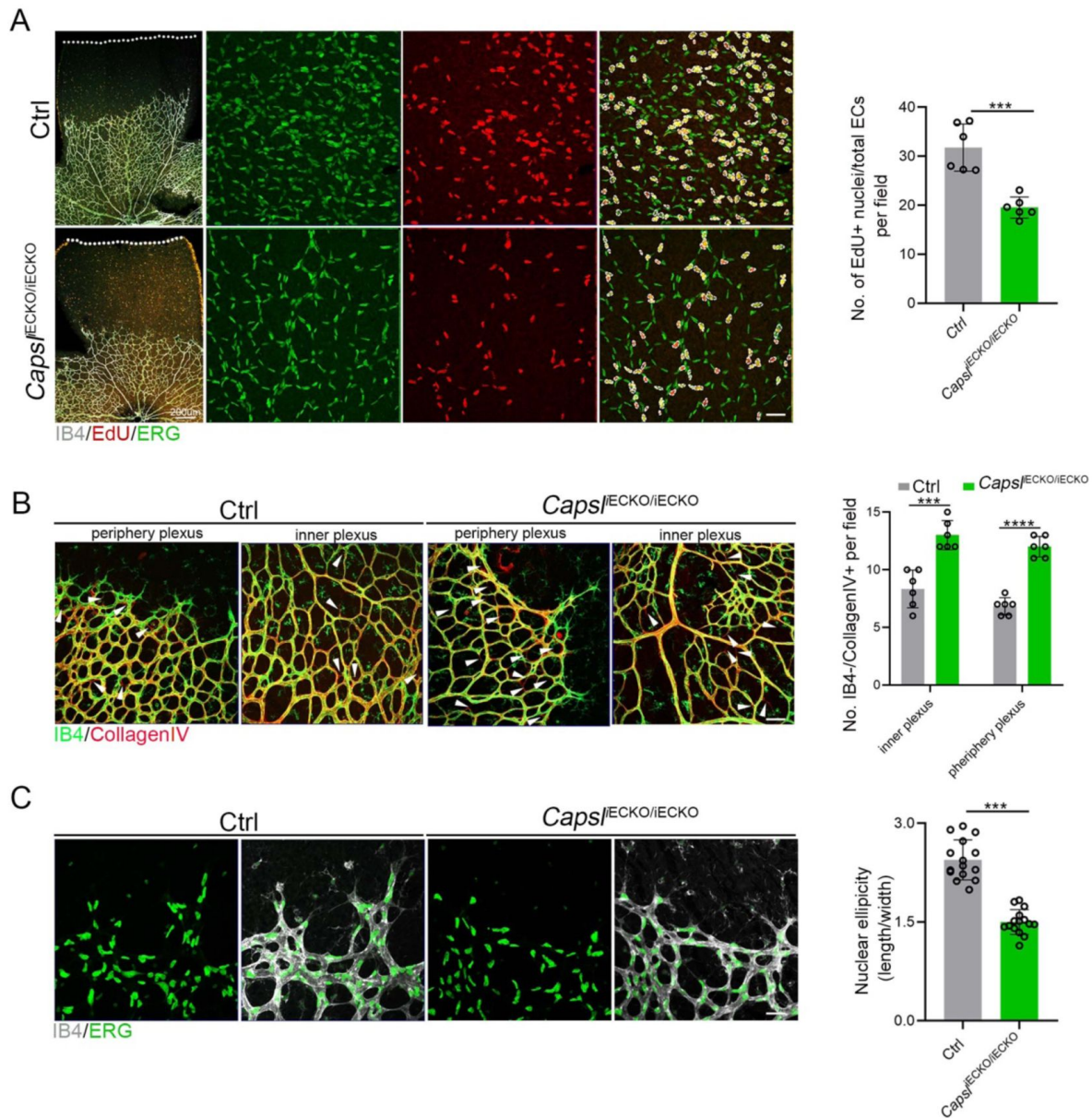


Figure 4.

Deletion of *Capsl* impairs EC proliferation and migration.

(A) Retina endothelial cell proliferation of Ctrl and *Capsl^{IECKO/IECKO}* mice at the vitreal surface was measured with EdU and ERG labeling at P5. Images captured at higher magnification are shown at right. Dashed lines mark the edge of the developing retina, and dashed circles represent both EdU+ and ERG+ cells. EC proliferation ability was measured by the ratio of EdU+ and ERG+ cells per vascular area. Scale bar: 200 μ m and 50 μ m (enlarged insets). Error bars indicate the SD. *** $P < 0.001$, by Student's *t* test ($n = 6$). **(B)** Representative images of retinal vessels at the periphery plexus and inner plexus of Ctrl and *Capsl^{IECKO/IECKO}* mice at P5 costained with IB4 (green) and Collagen IV (red). Arrowhead point to empty Collagen IV sleeves. And quantification of ratio of Collagen IV positive vessel segments to IB4 labeling-negative vessel segments. Scale bar: 50 μ m. Error bars indicate the SD. *** $P < 0.001$, **** $P < 0.0001$, by Student's *t* test ($n = 6$). **(C)** Magnified images of IB4+ vessels and ERG+ nuclei of ECs at angiogenic front of Ctrl and *Capsl^{IECKO/IECKO}* mice at P5. Quantification of tip cell nuclear ellipticity at the angiogenic growth front. Scale bar: 50 μ m. Error bars indicate the SD. *** $P < 0.001$, by Student's *t* test ($n = 14$).

the nuclei of actively migrating tip cells are elliptical and exhibit increased sphericity (decreased ellipticity) in static tip cells^[60,61]. In Ctrl mice retinæ, the nuclei of tip ECs were predominantly elliptical, pointing towards the avascular area (**Fig.4C**). Conversely, in the retinæ of *Capsl*^{iECKO/iECKO} mice, the nuclei of tip ECs appeared more spherical and did not orient towards the avascular area (**Fig.4C**). These findings suggest that CAPSL is critical for EC proliferation, vascular regression, and cell polarity.

Endothelial CAPSL is required for cell polarity and filopodia/lamellipodia formation

To investigate the functional mechanism and signaling cascade by which CAPSL regulates EC behaviors, we performed *in vitro* experiments on primary human retinal microvascular endothelial cells (HRECs). Stable CAPSL knockdown HRECs were generated using the lentivirus-mediated shRNA system. RT-qPCR analysis showed that the mRNA level of CAPSL was significantly decreased (Fig.S1E). In line with the overall reduced vascular coverage in *Capsl*^{iECKO/iECKO} mice retina, CAPSL-knockdown HRECs (shCAPSL-ECs) exhibited considerably impaired tube formation *in vitro* (**Fig.5A**). EdU labeling also revealed decreased cell proliferation of shCAPSL-ECs compared to that of shCtrl-ECs (**Fig.5B**). We further conducted flow cytometry to determine whether cell cycle arrest occurred upon depletion of CAPSL in HRECs. The redistribution of cell cycle phases indicated that the cell cycle was arrested during the G0/G1 phase in the shCAPSL-ECs (**Fig.5C**-D). To further investigate the role of CAPSL in EC migration, we performed the scratch-wound assay in confluent shCAPSL-ECs and shCtrl-ECs, as previously reported^[61,62]. Nine hours after scratching, ECs were co-stained with GM130 and phalloidin to visualize the Golgi apparatus and the cytoskeleton. The results indicated that the nuclei of shCAPSL-ECs were more spherical than those of shCtrl-ECs (**Fig.5E**-F). Given that the axial polarity can reflect the direction and migration of ECs^[63,64], we further measured the cell polarity (nucleus-to-Golgi polarity axis) of ECs at the wound edge^[61,62]. As a result, the Golgi apparatus of shCtrl-ECs was predominantly positioned around the wound edge, while that of shCAPSL-ECs showed more random positioning⁶⁵ (**Fig.5H**).

At the angiogenic front, the dynamic behaviors of tip cells largely rely on the rearrangement and organization of the actin cytoskeleton⁶⁶. Tip cells extend actin-driven filopodia and lamellipodia to probe chemotactic guidance cues, providing direction and migration for the developing vascular network^[67-69]. Notably, shCAPSL-ECs exhibited fewer filopodia, smaller lamellipodia, and significantly impaired wound closure compared to shCtrl-ECs at the wound scratch edge (**Fig.5I**). Also, as expected, shCAPSL-ECs exhibited significantly impaired wound closure at the same time point after scratching relative to shCtrl-ECs (**Fig.5J**). Small Rho GTPases, including CDC42 and Rac, are essential for lamellipodial extension and cell migration^[70,71]. Western blot analysis uncovered markedly attenuated expression of CDC42, RHOA, and Rac1 (**Fig.5K**). Moreover, both MYL9 and phosphorylated MYL9 at Ser19, which are crucial regulators of contractile force, were also drastically reduced in shCAPSL-ECs compared to those of shCtrl-ECs (**Fig.5K**). These results indicated that CAPSL regulates EC polarity, filopodia/lamellipodia formation, and migration by modulating the actomyosin cytoskeleton.

CAPSL regulates endothelial cell proliferation through the MYC signaling axis

Given the fact that FEVR is predominantly characterized by the compromised Norrin/ β -catenin signaling pathway as its main causative factor^[10,15,17,18], we first asked whether loss of CAPSL function causes FEVR through Norrin/ β -catenin signaling. TopFlash reporter gene system was used to determine the activity of the Norrin/ β -catenin signaling pathway. Knocking down CAPSL expression in human embryonic kidney 293 (HEK293) SuperTopFlash (STF) cells led to similar luciferase activity, compared to cells transfected with shCtrl (Fig.S3A). Furthermore, transfection of shRNA-resistant CAPSL plasmid (WT, R30X, or L83F) in CAPSL-knockdown 293STF

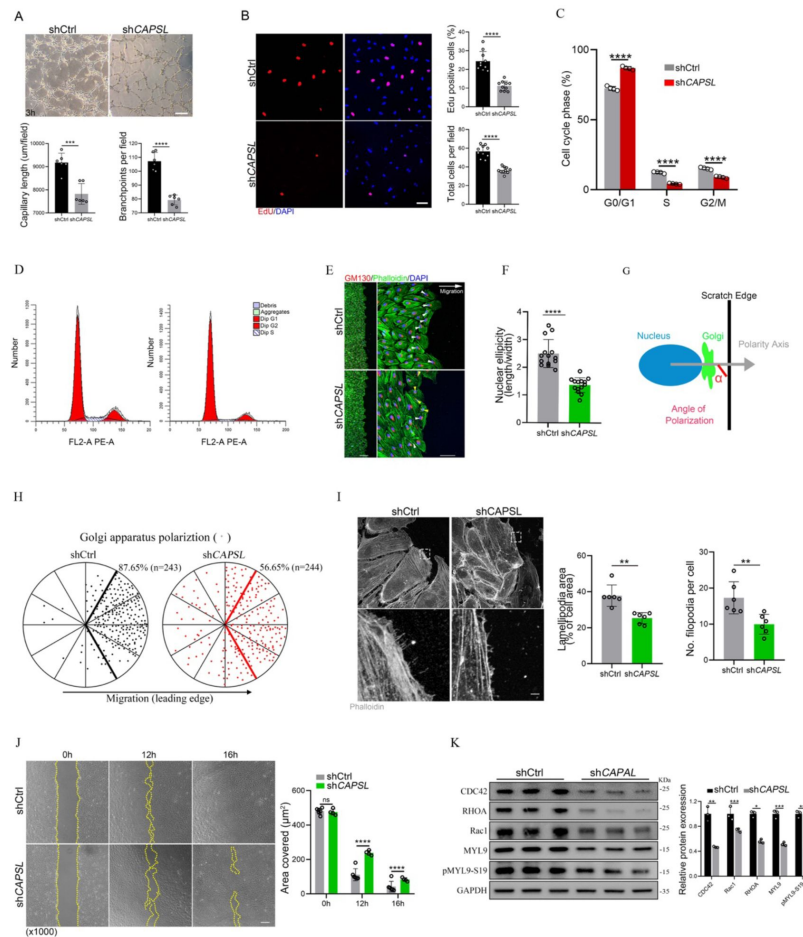


Figure 5.

Depletion of CAPSL in HRECs compromises *in vitro* EC proliferation and migration.

(A) Representative images of *in vitro* tube formation after transfection of HRECs with shRNA. Scale bar: 200 μ m. Error bars indicate the SD. *** P < 0.001, **** P < 0.0001, by Student's *t* test (n =6). **(B)** Incorporation of EdU in shRNA transfected HRECs. Representative confocal images and quantification of proliferating HRECs both in number per field and proportion of EdU-positive cells. Scale bar: 50 μ m. Error bars indicate the SD. **** P < 0.0001, by Student's *t* test (n =10). **(C-D)** Cell cycle analysis of shCtrl-ECs and shCAPSL-ECs by flow cytometry. Error bars indicate the SD. **** P < 0.0001, by Student's *t* test (n =4). **(E)** Representative images of phalloidin actin cytoskeleton (green) and GM130 (red) showing polarity angles of shCtrl-ECs and shCAPSL-ECs at the edge of scratch wound. The arrow points toward the wound. Colored arrowheads represent different migration state. Scale bar: 200 μ m (left panel) 50 μ m (right panel). **(F)** Quantification of nuclear ellipticity of HRECs at the margin of wound scratch. Error bars indicate the SD. **** P < 0.0001, by Student's *t* test (n = 14). **(G)** Schematic pictures showing the define of polarity axis of each cell. Polarity axis was measured with the angle (α) between the scratch edge and the vector drawn from the center of nucleus to the center of the Golgi apparatus. **(H)** Polar plots showing Golgi apparatus polarization. The bold lines represent 120° region centered on the vector, which is perpendicular to the wound scratch. The dots represent the angle (α) of each cell and the numbers indicate the frequency of dots within the 120° region of the bold line of shCtrl-ECs (n =243) and shCAPSL-ECs (n =244). **(I)** Images of phalloidin-stained actin cytoskeleton and comparisons of indicated parameters in shCtrl-ECs and shCAPSL-ECs at the edge of scratch wound. The dashed boxed region is shown at higher magnification at the bottom panel. Scale bar: 50 μ m (top panels), 25 μ m (bottom panels). **(J)** Representative images of wound scratch assay at 0h, 12h, and 16h after wound was made. And the quantification of covered area at different time point. The dashed line indicates the gap of the wound after wound scratch at different time point. Scale bar: 200 μ m. Error bars indicate the SD. **** P < 0.0001, ns< no significance, by Student's *t* test (n = 4). **(K)** Immunoblot and quantification analysis of expression of small GTPase proteins and a key regulator of contractile force MYL9 in shCtrl-ECs and shCAPSL-ECs. Error bars indicate the SD. * P < 0.05, ** P < 0.01, *** P < 0.001, by Student's *t* test (n = 3).

cells resulted in no significant difference in luciferase activity (Fig.S3B). These results indicated that *CAPSL* was not a major player in Norrin/ β -catenin signaling to regulate retinal vascular development.

We next performed unbiased transcriptomic and proteomic analyses on the control and *CAPSL*-knockdown HRECs to explore the underlying mechanism by which *CAPSL* regulates EC proliferation and migration. RNA sequencing (RNA-seq) identified 1961 up-regulated genes and 2205 down-regulated genes in sh*CAPSL*-ECs compared to shCtrl-ECs (logFC>0.58, Pvalue<0.05), 1342 up-regulated genes and 1218 down-regulated genes in ECs overexpressing *CAPSL* (LentiOE-ECs) compared to Ctrl-ECs (Fig.6A). Gene set enrichment analysis (GSEA) on transcriptomic data showed multiple down-regulated signaling axes upon the depletion of *CAPSL*, including E2F targets (NES=-3.2944438, P value=0), G2/M checkpoints (NES=-3.2851038, Pvalue=0), MYC targets (NES=-2.8795638, P value=0), and mitotic spindle (NES=-2.646523, Pvalue=0) (Fig.6B-C). In line with this, overexpression of *CAPSL* in HRECs also caused a significant increase in E2F targets (NES=2.7765026, P value=0), G2/M checkpoints (NES=2.6528199, Pvalue=0), and MYC targets (NES=2.6528199, Pvalue=0) (Fig.6D-E). Meanwhile, proteomic profiling showed similar expression patterns with that of transcriptomic profiling upon knockdown or overexpression of *CAPSL* (Fig.6F-I). Furthermore, we applied correlation analysis on the transcriptomic and proteomic data, which divided the genes into 9 quadrants (Fig. 6J). GSEA analysis was performed on genes in the third and seventh quadrants, which were regulated at the transcriptional level (Fig. 6K). Interestingly, the down-regulated signaling pathways are highly consistent with those observed in GSEA analysis of both transcriptomic and proteomic analyses (Fig.6K). Given that the c-MYC was an established driver of cell growth and migration⁷², and that the genes in the MYC signaling were down-regulated upon *CAPSL*-depletion (Fig.6L), we speculated that *CAPSL* might play a pivotal role in MYC signaling in HRECs.

We next incorporated an analysis of transcriptome profiling of human umbilical vein endothelial cells (HUVECs) infected with lentiviruses encoding control-or c-MYC-targeting gRNAs (GSE161815)⁷³. Intriguingly, GSEA analysis revealed that ablation of cMYC in HUVECs led to compromised E2F targets, MYC targets, and G2/M checkpoint signaling activity (Fig.7A-B), largely consistent with those observed in the absence of *CAPSL*. This also suggests that cMYC might play a role in the upstream regulation of E2F targets and G2/M checkpoint signaling. Venn diagram analysis of down-regulated genes (LogFC<-0.585, P<0.05) in *CAPSL*-depleted HRECs and c-MYC-depleted HUVECs revealed a large proportion of shared genes, indicative of similar transcriptional regulatory patterns between *CAPSL* and cMYC (Fig.7C). Additionally, these genes are predominantly enriched in the MYC targets signaling (Fig.7D-F).

We, therefore, asked whether *CAPSL* regulates c-MYC expression. Notably, the depletion of *CAPSL* significantly decreased the c-MYC protein level, rather than the mRNA level in HRECs (Fig.7G). Subsequently, western blot analysis revealed that expression levels of c-MYC targets, such as MCM family, E2F1, CyclinD family, and PCNA, were decreased in the absence of *CAPSL* in HRECs (Fig.7H-I). Taken together, these findings suggest c-MYC as a potential functional regulator downstream of *CAPSL*.

Discussion

Over the last decades, major progress has been made in understanding the molecular mechanisms underlying FEVR, and 17 FEVR-causing genes have been identified^[10,31]. However, the pathogenic genes and mechanisms underlying approximately 50% of clinical cases remain elusive^[20,21]. In this study, we report a new FEVR candidate gene *CAPSL* and uncover the pivotal roles of *CAPSL* in retina vascular development. The EC-specific *Capsl*-knockout mouse model exhibited FEVR phenotypes, including delayed vascular progression, retarded hyaloid

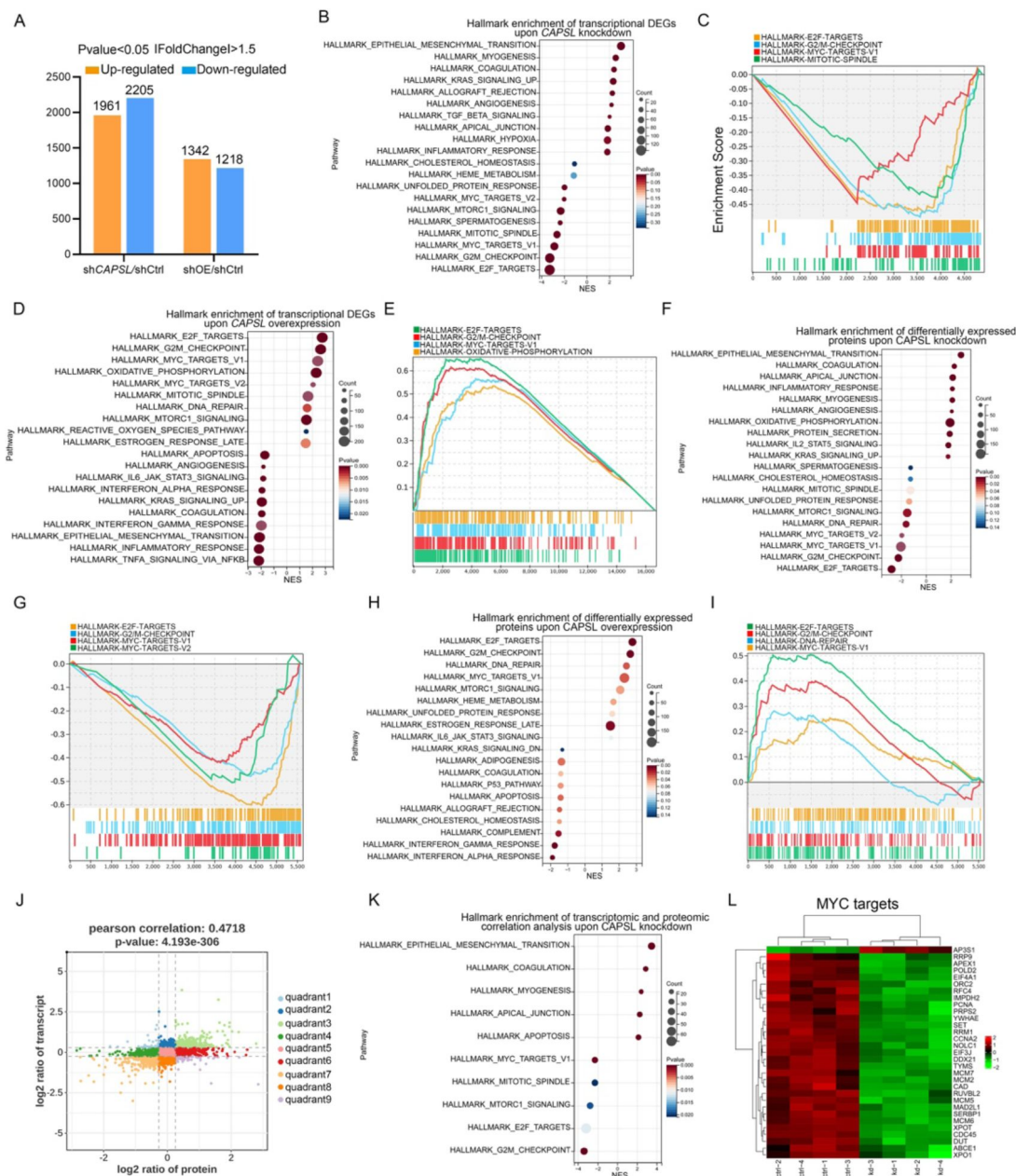


Figure 6.

CAPSL suppresses MYC signaling axis.

(A) Differential gene expression information of shCAPSL-ECs versus shCtrl-ECs group and LentiOE-ECs versus shCtrl-ECs group. (B-E) Gene set enrichment analysis (GSEA) analysis on the RNA sequencing data of HRECs. Top 10 ranked up or down regulated signaling axis were listed (B) and top 4 down-regulated gene sets were listed (C) in comparison of shCtrl-ECs versus shCAPSL-ECs. Top 10 ranked up or down regulated signaling axis were listed (D) and top 4 up-regulated gene sets were listed (E) in comparison of LentiOE-ECs versus shCtrl-ECs. (F-I) Gene set enrichment analysis (GSEA) analysis on the proteomic profiling data of HRECs. Top 10 ranked up or down regulated signaling axis were listed (F) and top 4 down-regulated gene sets were listed (G) in comparison of shCtrl-ECs versus shCAPSL-ECs. Top 10 ranked up or down regulated signaling axis were listed (H) and top 4 up-regulated gene sets were listed (I) in comparison of LentiOE-ECs versus shCtrl-ECs. (J) Correlated RNAs and proteins enriched in nine quadrants of shCtrl-ECs versus shCAPSL-ECs. (K) Gene set enrichment analysis (GSEA) analysis on the genes/proteins in quadrants 3 and 7, and top 5 ranked up or down regulated signaling axis were listed. (L) Clustered heat map of the expression fold changes of several MYC signature genes of in both RNA profiling and proteomic profiling of shCtrl-ECs and shCAPSL-ECs.

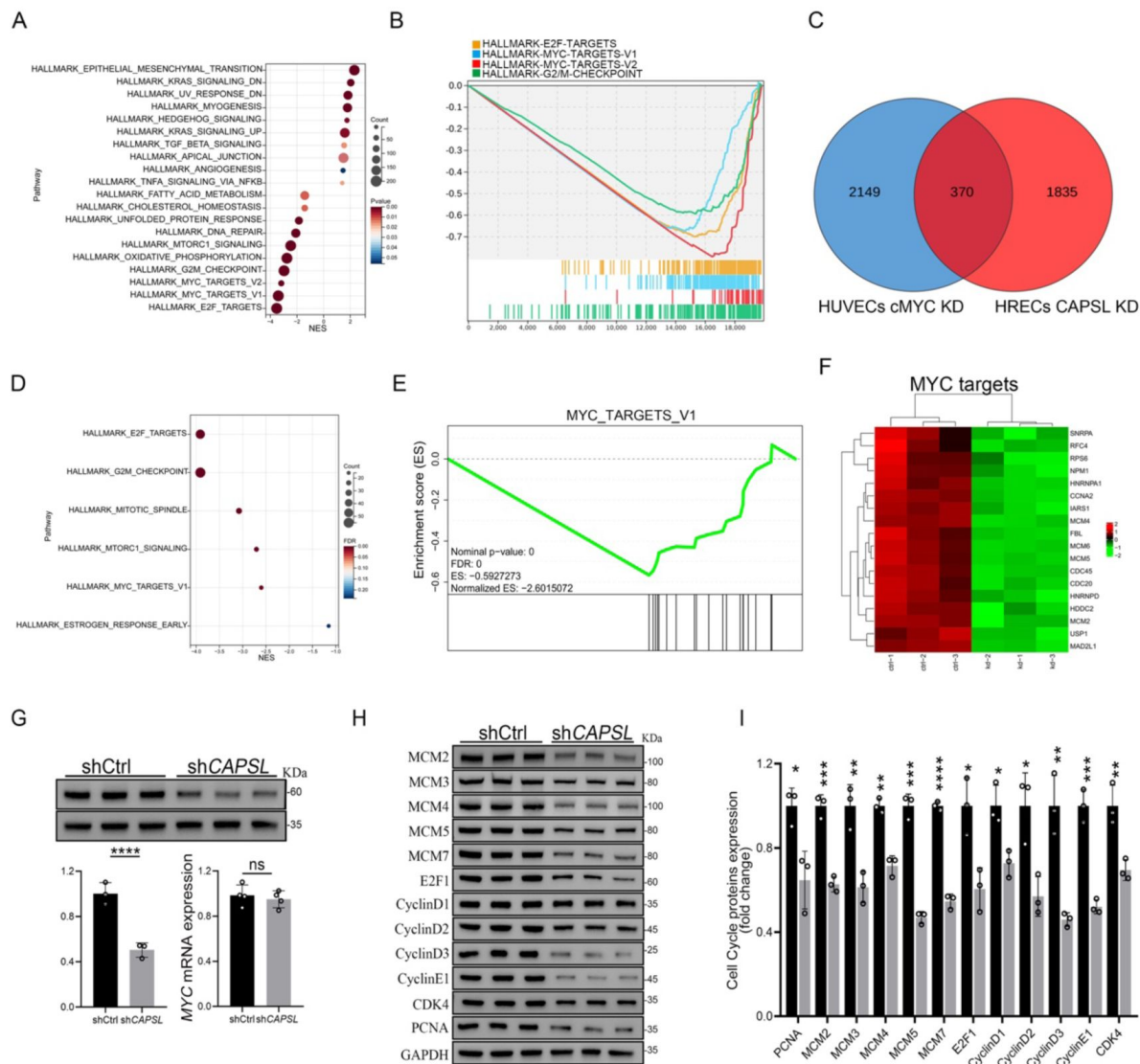


Figure 7.

Loss of *CAPSL* led to similar transcriptional regulatory patterns to the loss of *c-MYC* in HUVECs.

(A-B) Gene set enrichment analysis (GSEA) analysis on the RNA sequencing data in comparison of shCtrl-HUVECs versus shc-MYC-HUVECs. Top 10 ranked up or down regulated signaling axis were listed **(A)** and top 4 down-regulated gene sets were listed **(B)**. **(C)** Venn diagram analysis of down-regulated genes in *CAPSL*-depleted HRECs and *c-MYC*-depleted HUVECs. **(D-E)** Gene set enrichment analysis (GSEA) analysis on the shared down-regulated genes in *CAPSL*-depleted HRECs and *c-MYC*-depleted HUVECs. **(F)** Heatmap of MYC signature genes of shared genes based on Venn analysis. **(G)** *c-MYC* expression level in shCtrl-ECs and shCAPSL-ECs was quantified by western blot and RT-qPCR. Error bars indicate the SD. **** $P < 0.0001$, ns: no significance by Student's *t* test ($n = 3$). **(H)** Western blot analysis of expression of MYC targets. **(I)** Quantification analysis of MYC targets. Error bars indicate the SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, by Student's *t* test ($n = 3$)

vessel regression, and decreased vessel density. Deficiency of CAPSL in ECs resulted in compromised cell proliferation and defective EC migration, providing insights into the regulatory roles of CAPSL in retina vascularization (**Fig.2**–5).

During retinal development, angiogenesis comprises various processes such as endothelial cell (EC) specification, adhesion, proliferation, migration, and pruning⁴⁷. The regulation of angiogenesis entails the control of EC behaviors and the interactions between ECs, extracellular matrix (ECM), and other types of retinal cells⁶⁵. In this study, we observed impaired radial expansion and vertical invasion, as well as the increased pruning of the retinal vascular ECs upon inactivating endothelial *Capsl* (**Fig.2**–3). However, neither the expression level nor the gradient of VEGFA was disturbed in *Capsl*^{IECKO/IECKO} mice, which could be attributed to the intact blood barrier in the *Capsl*^{IECKO/IECKO} retina (Fig.S2). In addition, the loss of *Capsl* resulted in the presence of more spherical tip cell nuclei, which were directed more randomly to the avascular area (**Fig.4**). These results provide evidence that CAPSL acts as a novel regulator for collective EC behavior.

In vitro functional studies demonstrated that depletion of CAPSL impaired tube formation, EC proliferation ability, and EC polarity (**Fig.5**). Moreover, the formation of filopodia and lamellipodia is also disturbed by the depletion of CAPSL (**Fig.5**). Using the unbiased transcriptomic and proteomic sequencing, bioinformatic analysis, and western blot assay, we demonstrated that the defects in CAPSL affect EC function by down-regulating the MYC signaling cascade (**Fig.6**–7). c-MYC has been shown to be a critical mediator of anabolic metabolism, cell growth, and migration^[74,75]. Aberrant upregulation of the MYC signaling pathway is frequently observed in many types of cancers^[76,77]. Interestingly, the phenotypes of retinal vessels in *Capsl*^{IECKO/IECKO} mice resemble the vascular defects in *c-Myc*^{IECKO/IECKO} mice, exhibiting FEVR-like phenotypes, such as impaired vascular progression, decreased branch points, and reduced EC proliferation^[39,42].

The Norrin/ β -catenin signaling pathway is a subset of Wnt signaling pathway that is specifically activated during vascular development of the central nervous system^[20,78]. Although several signaling pathways have been implicated in the pathogenesis of FEVR, variants in crucial components of Norrin/ β -catenin signaling account for most of the FEVR cases^[15,17,18]. It is worth mentioning that c-MYC is widely recognized as a downstream target of Wnt signaling, which is transcriptionally regulated during Wnt activation^[79–81]. However, to our knowledge, there is a lack of literature reporting on the correlation between the MYC signaling pathway and FEVR. Interestingly, here we proved that the absence of CAPSL leads to a decrease in c-MYC protein level, which down-regulates MYC signaling without affecting Norrin/ β -catenin signaling (**Fig.7** and S3). Consequently, we propose c-MYC as a potential therapeutic target for FEVR, a hypothesis that warrants substantiation through subsequent investigations.

Additionally, the formation of filopodia and lamellipodia, which is reported to be intimately related to the actomyosin dynamics and the activity of small Rho GTPases such as CDC42^[82,83], is significantly reduced in CAPSL-defective HRECs. In our study, we confirmed that the knockdown of CAPSL led to a significant reduction of CDC42, which might be the causative mechanism behind the disturbed formation of filopodia and lamellipodia (**Fig.5**).

Taken together, our study demonstrated that CAPSL is potentially a novel candidate gene associated with FEVR. We also demonstrated that variants in the CAPSL gene, independently of canonical Norrin/ β -catenin signaling, cause FEVR through inactivating MYC signaling, expanding FEVR-involved signaling pathway and providing a potential therapeutic target for the intervention of FEVR.

Methods

DNA sequencing of patients and controls

The study was approved by the Institutional Research Committees of Sichuan Provincial People's Hospital and Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine. Informed consent was obtained from all participants and from guardians of minors involved in genetic testing. The patients were clinically diagnosed with exudative vitreoretinopathy. Genomic DNA was extracted from the peripheral blood of FEVR families and control individuals using a blood DNA extraction kit (Qiagen, Germantown, Maryland, USA) according to standard protocol. Whole-exome sequencing (WES) was performed on DNA samples to identify candidate pathogenic genes for FEVR as previously described¹⁷. Family ID numbers 3036 and 3104 are not known to anyone outside the research group.

Mouse model and genotyping

All mice used in the study were on a C57BL/6J genetic background, and *Capsl*^{loxP/+} mice were bred with *Pdgfb-iCreER*⁴³ transgenic mice to generate *Capsl*^{loxP/loxP}, *Pdgfb-iCreER* mice. Genomic DNA samples extracted from mouse tails were genotyped using PCR to detect loxP-flanked *Capsl* and *Pdgfb-iCreER*. Primer sets used for loxP were: 5'-GGCAGGTAAGATGGTGTC-3' and 5'-TCTGTTTGTGGATCAATGTG-3'; and primer sets used for *Pdgfb-iCreER* were: 5'-GCCGCCGGGATCACTCTCG-3' and 5'-CCAGCCGCCGTCGCAACTC-3'. All mice were maintained under specific pathogen-free conditions with standard rodent water and diet, both males and female mice were used for the analysis. Experiments involving animals were conducted in accordance with institutional guidelines and following the protocols approved by the Institutional Animal Care and Use Committee of Sichuan Provincial People's Hospital.

Preparation of flat-mount retinæ and retina sections

Enucleated eyes were fixed for 15 min in 4% paraformaldehyde (PFA) at room temperature, followed by immersion in PBS for the same duration. The retinæ were then dissected and partially cut into four quadrants as previously described⁸⁴. The dissected retinal cups were post-fixed overnight and then stained as described below. As for retina sections, after fixation in 4% PFA for 2h, the enucleated eyes were dehydrated until they settled to the bottom of the tube with 30% sucrose and then embedded in Tissue Freezing Medium (Servicebio). Specimens were sectioned, cleaned, and stained as previously described⁸⁵.

Hyloaid vessel preparation

Neonatal eyes were fixed in 4% PFA for 2h before the cornea, lens, and iris were removed. The eye cup was immersed in 5% gelatin at 4 °C overnight. The solidified gelatin was extricated and heat melted on a glass slide, washed with warm water, air-dried, and imaged with DAPI staining.

Immunohistochemistry

All immunostainings of retinæ were performed using littermates with similar body size and treated under the same conditions. The flat-mount retina and retina sections were permeabilized and blocked in PBS containing 5% fetal bovine serum and 0.2% Triton X-100 at room temperature for 2h. Next, they were rinsed in PBS and incubated in directly conjugated Isolectin-B4 Alexa 488 or Alexa 594 (1:200; Thermo Fisher, USA) and primary antibodies in a blocking buffer at 4°C overnight. After three washes, the retinæ were processed for multiple labeling or flat-mounted onto microscope glass slides with Fluoromount (Sigma-Aldrich, USA).

For *in vivo* analysis of cell proliferation, each pup was injected intraperitoneally with 200ug EdU 3h before the mice were euthanasia. Retinae were dissected and permeabilized as previously described. EDU-positive labeling was stained and detected by means of a Click-iT EDU Alexa Fluor-647 Imaging Kit (C10640, Thermo Fisher Scientific, USA) according to the manufacturer's instruction. For paracellular BRB integrity analysis, each pup was injected intraperitoneally with 2% fluorescent tracer, 5-(and-6)-tetramethylrhodamine biocytin, Biocytin-TMR (CAT#T12921, Thermo Fisher Scientific, USA) for 24 hours prior to sacrifice, followed by flat-mount preparation, blocking, and staining of Isolectin B4.

Images of stained flat-mounted retinae, retina sections, and HRECs were acquired using Zeiss LSM 800 confocal microscope (Thornwood, NY, USA) and processed with Zeiss processing software, Angio Tool, and Adobe Photoshop. The detailed information about antibodies used for immunofluorescence is listed in Table S2.

Cell lines and primary cell culture

Human retinal microvascular endothelial cells (HRECs) were purchased from Cell System and cultured in EBM™-2 media (CC-3156, Lonza, Switzerland) containing 5% fetal bovine serum, 0.4% hydrocortisone, 4% Hfgf-b, 0.1% VEGF, 0.1% R3-IGF, 0.1% ascorbic acid, 0.1% hEGF 0.1% GA-1000, and 0.1% heparin at 37°C in a 5% CO₂ incubator. HRECs at passages 3-7 were used for experiments.

HEK-293T were obtained from the American Type Culture Collection (ATCC) and cultured in DMEM (SH30023.01, Hyclon, USA) with 10% fetal bovine serum and cells were maintained at 37°C in a 5% CO₂ incubator.

Gene knockdown and overexpression strategies

HRECs were seeded on 6 cm dishes at the day before transfected with lentivirus carrying shRNA for human *CAPSL* (5'-AAGACCTTCGTGAAGTATA-3', Genechem, Shanghai, China) or negative control shRNA (5'-TTCTCCGAACGTGTCACGT-3') according to the protocol of the manufacture. Cells were incubated for at least 72h before used for further experiments.

For overexpression of exogenous protein in HEK-293T cells, cells were transfected with Lipofectamine-3000 (Invitrogen, USA) according to the manufacturer's instructions.

Matrigel EC tube formation assay

HRECs transfected with the corresponding shRNA after 72h were counted and planted onto the matrigel as previously described⁸⁶. After 6h of incubation at 37°C in a 5% CO₂ incubator, the images were captured under an anatomical lens (Carl Zeiss, Germany).

Immunofluorescence of HREC

After cell counting, the same number of HRECs transfected with shCtrl and shCAPSL were seeded on slides in 24-well plates. Cells were fixed with 4% PFA at room temperature for 15min. After being rinsed in PBS three times, cells were blocked in the blocking buffer at room temperature for 2h. The cells were labeled with primary antibodies overnight at 4°C. Following three washes in PBS, Alexa-Fluor-594 or Alexa-Fluor-647 conjugated antibodies were administered along with DAPI, and cells were incubated at room temperature for 1h. After three washes in PBS, stained cells were mounted on microscope glass slides with Fluoromount (Sigma-Aldrich, USA).

For *in vitro* analysis of cell proliferation, 20 μM EdU (Thermo Fisher Scientific, USA) was added into 96-well plated 3h before cells were harvested. EdU-positive cells were stained and detected with the Click-iT EDU Alexa Fluor-647 Imaging Kit (C10640, Thermo Fisher Scientific, USA) according to the manufacturer's instruction.

Wound healing assay

72h after shRNA transfection, HRECs were seeded in 6-well plates with complete medium for 24h until grown to confluence. Plates were then coated with 10 µg/mL fibronectin for half an hour in 37 °C before cell culture. 200 µL pipette tips were used to generate wounds, and each well was washed twice with Utrasalin A (Lonza, Switzerland) to remove detached cells. Cells were then starved in the EBM-2 medium at 37°C in a 5% CO₂ incubator. Images of the wounds were captured with an optical microscope at 0h, 12h, and 16h after the wounds were made, respectively. The wound closure areas were measured using ImageJ software.

For analyses of Golgi apparatus polarization, cells were maintained on 24-well glass slides and fixed with 4% PFA for 15 min at 9 h after cell migration was initiated.

RNA extraction and RT-qPCR

Total RNA was extracted from cultured HRECs or fresh lungs from mouse using TRIzol reagent (Life Technologies, USA) according to the manufacturer's instructions. RNA concentration and quality were measured with NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA), and 1 µg total RNA was reverse transcribed using EasyScript One-Step RT-PCR SuperMix (TransGen Biotech, Beijing, China) following the standard protocol. RT-qPCR was performed using a 7500 Real-Time PCR System (Applied Biosystems, USA) with TransStart Tip Green qPCR Supermix (TransGen Biotech). At least three experiments were performed, and PCR reactions were performed in triplicate. GAPDH was used as a reference gene. The primer sets are listed in Table S3.

Protein extraction and western blotting

HRECs or lung tissue from mice were lysed in standard PIPA lysis buffer (50 mM Tris-HCl, 150 mM NaCl, 1% Triton X-100, 0.5% sodium deoxycholate, 0.1% sodium dodecyl sulfate, pH 7.4) supplemented with Protease and Phosphatase Inhibitor Cocktail (Roche, USA). After sonication, centrifugation, and protein quantification, the supernatant was diluted in 2x SDS loading buffer. Equal amounts of protein (20 µg) were loaded and resolved on 10-15% SDS-polyacrylamide, and Western blotting was performed. After gels were transferred onto a Polubinylidene Difluoride (PVDF) membrane (GE Healthcare, USA), the membranes were incubated in a blocking buffer containing 8% skimmed milk (9999, Cell Signaling Technology) in TBST for 1h at room temperature on a rocking platform, followed by overnight incubation at 4°C with the primary antibodies diluted in blocking buffer. Membranes were then washed in TBST three times and incubated in the blocking buffer containing secondary HRP-conjugated antibodies for 1h at room temperature. Protein blots were visualized with e-BLOT Touch Imager (e-BLOT, China) and blots were relatively quantified with ImageJ. GAPDH was used as an internal reference. Detailed information about antibodies used for western blot is listed in Table S4.

Luciferase Assays

Dual-luciferase assays were performed as previously described¹⁷. Briefly, 293STF cells were plated on 24-well plates at a confluency of ~30%. The following day, the cells were transfected with corresponding plasmids with lip3000. 48 hours post-transfection, cells were harvested, and the lysates were used to measure Firefly and *Renilla* luciferase activity according to the manufacturer's instructions (TransGen).

Quantification of retinal parameters

All quantifications were done using Zeiss processing software on high-resolution confocal images. Vascular progression was measured in a straight line from the optic nerve to the angiogenic front of the retinal plexus for each retina quadrant (n=6, each group). Vessel density and branch points

were calculated with Angio Tool software from flat-mounted retinæ (n=6, each group). Endothelial tip cell numbers were measured by counting endothelial sprouts at the angiogenic front of the entire vascular plexus of the same length (n=6, each group). Endothelial tip cell filopodia density was calculated with high-resolution images (60× objective) of 6 randomly selected angiogenic front fields from 6 retinæ per genotype (n=6, each group). Endothelial cell proliferation was calculated by measuring the number of both EDU-positive and ERG-positive cells within retinæ of the same size and same position at the edge of the vascular plexus (n=6, each group). Retinal pericyte coverage was quantified by comparing the NG2⁺ or Desmin⁺ pericytes with IB4⁺ vessels (n=6, each group). The nuclear ellipticity of the cells at the scratch edge was calculated by the distance of the nuclear long axis (Height) divided by the maximum vertical distance to the nuclear long axis (Width).

Transcriptomic profiling

Total RNA was extracted from shCtrl-ECs, shCAPSL-ECs, and LentiOE-ECs and quantified using NanoDrop2000 Spectrophotometer (Thermo Fisher Scientific, USA). RNA integrity was determined by Bioanalyzer RNA 6000 Nano assay kit (Agilent, China). RNA library construction was performed with Illumina Truseq™ RNA sample prep Kit and RNA sequencing was performed with Truseq SBS Kit (300 cycles) in BIOZERON (BIOZERON, China). RNA-Seq reads were performed using the TopHat software tool to acquire the alignment file, which was used to quantify mRNA expression and determine the differentially expressed genes. All gene expression values were changed to log2 values for further analysis, and a P value of less than or equal to 0.05 was considered to indicate significance. The gene set enrichment analysis (GSEA) was performed with the omicshare platform (<https://www.omicshare.com> .

Proteomic profiling

Total protein was extracted from shCtrl-ECs, shCAPSL-ECs, and LentiOE-ECs and quantified using a BCA protein quantification kit (Beyotime, China). Samples were sent to PTM Biolab (PTM Biolab, China) to perform proteomic profiling using 4D-Labelfree. All protein expression values were changed to log2 values for further analysis, with a corrected P value < 0.05. The gene set enrichment analysis (GSEA) was performed with the omicshare platform (<https://www.omicshare.com> .

Image acquisition and statistical analysis

Immunofluorescence images were obtained using a laser scanning microscope (LSM 900, Zeiss). Statistical analyses were performed with GraphPad Prism 8.0 software. Data were analyzed using the Student's t-test and plotted as mean ± SD. At least three independent experiments were performed. Image J was used to compare results for the tube formation assay experiments and bands of immunoblot. Results were considered significant when *P<0.05, **P<0.01, ***P<0.001, or ****P<0.0001. “ns” stands for “no significance.”

Study approval

This study was approved by the institutional review board of Sichuan Provincial People's Hospital (approval number: 2014-009) and Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine, and informed consent was obtained from all participants or legal guardians for minors. All animal studies were performed according to established ethical guidelines approved by the animal care committee of Sichuan Provincial People's Hospital.

Competing interests

All authors declare that there are no competing interests.

Author contributions

XZ, ZY, and PZ conceived the research. WL, SL, MY, and JM performed the animal mouse model research and cell biology, transcriptome, and proteome data analysis. LL, PF, and QX performed animal breeding and genotyping. PZ recruited the participants and performed the WES study. LH performed the Sanger sequencing analysis. WL, SL, MY, and XZ wrote the manuscript, with input from other authors.

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Data Availability

All data produced in the present study are available upon reasonable request to the authors.

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Editors

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Reviewer #1 (Public Review):

Summary:

The author presents the discovery and characterization of CAPSL as a potential gene linked to Familial Exudative Vitreoretinopathy (FEVR), identifying one nonsense and one missense mutation within CAPSL in two distinct patient families afflicted by FEVR. Cell transfection assays suggest that the missense mutation adversely affects protein levels when overexpressed in cell cultures. Furthermore, conditionally knocking out CAPSL in vascular endothelial cells leads to compromised vascular development. The suppression of CAPSL in human retinal microvascular endothelial cells results in hindered tube formation, a decrease in cell proliferation, and disrupted cell polarity. Additionally, transcriptomic and proteomic profiling of these cells indicates alterations in the MYC pathway.

Strengths:

The study is nicely designed with a combination of in vivo and in vitro approaches, and the experimental results are good quality.

Weaknesses:

My reservations lie with the main assertion that CAPSL is associated with FEVR, as the genetic evidence from human studies appears relatively weak. Further careful examination of human genetics evidence in both patient cohorts and the general population will help to clarify. In light of human genetics, more caution needs to be exercised when interpreting results from mice and cell models and how is it related to the human patient phenotype.

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Reviewer #2 (Public Review):

Summary:

This work identifies two variants in CAPSL in two-generation familial exudative vitreoretinopathy (FEVR) pedigrees, and using a knockout mouse model, they link CAPSL to retinal vascular development and endothelial proliferation. Together, these findings suggest that the identified variants may be causative and that CAPSL is a new FEVR-associated gene.

Strengths:

The authors' data provides compelling evidence that loss of the poorly understood protein CAPSL can lead to reduced endothelial proliferation in mouse retina and suppression of MYC signaling in vitro, consistent with the disease seen in FEVR patients. The study is important, providing new potential targets and mechanisms for this poorly understood disease. The paper is clearly written, and the data generally support the author's hypotheses.

Weaknesses:

(1) Both pedigrees described appear to suggest that heterozygosity is sufficient to cause disease, but authors have not explored the phenotype of Capsl heterozygous mice. Do these animals have reduced angiogenesis similar to KOs? Furthermore, while the p.R30X variant protein does not appear to be expressed in vitro, a substantial amount of p.L83F was detectable by western blot and appeared to be at the normal molecular weight. Given that the full knockout mouse phenotype is comparatively mild, it is unclear whether this modest reduction in protein expression would be sufficient to cause FEVR - especially as the affected individuals still have one healthy copy of the gene. Additional studies are needed to determine if these variants alter protein trafficking or localization in addition to expression, and if they can act in a dominant negative fashion.

(2) The manuscript nicely shows that loss of CAPSL leads to suppressed MYC signaling in vitro. However, given that endothelial MYC is regulated by numerous pathways and proteins, including FOXO1, VEGFR2, ERK, and Notch, and reduced MYC signaling is generally associated with reduced endothelial proliferation, this finding provides little insight into the mechanism of CAPSL in regulating endothelial proliferation. It would be helpful to explore the status of these other pathways in knockdown cells but as the authors provide only GSEA results and not the underlying data behind their RNA seq results, it is difficult for the reader to understand the full phenotype. Volcano plots or similar representations of the underlying expression data in Figures 6 and 7 as well as supplemental datasets showing the differentially regulated genes should be included. In addition, while the paper beautifully characterizes the delayed retinal angiogenesis phenotype in CAPSL knockout mice, the authors do not return to that model to confirm their in vitro findings.

(3) In Figure S2D, the result of this vascular leak experiment is unconvincing as no dye can be seen in the vessels. What are the kinetics for biocytin tracers to enter the bloodstream after IP

injection? Why did the authors choose the IP instead of the IV route for this experiment? Differences in the uptake of the eye after IP injection could confound the results, especially in the context of a model with vascular dysfunction as here.

(4) In Figure 5, it is unclear how filipodia and tip cells were identified and selected for quantification. The panels do not include nuclear or tip cell-specific markers that would allow quantification of individual tip cells, and in Figure 5C it appears that some filipodia are not highlighted in the mutant panel.

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Reviewer #3 (Public Review):

Summary:

This manuscript by Liu et al. presents a case that CAPSL mutations are a cause of familial exudative vitreoretinopathy (FEVR). Attention was initially focused on the CAPSL gene from whole exome sequence analysis of two small families. The follow-up analyses included studies in which CAPSL was manipulated in endothelial cells of mice and multiple iterations of molecular and cellular analyses. Together, the data show that CAPSL influences endothelial cell proliferation and migration. Molecularly, transcriptomic and proteomic analyses suggest that CAPSL influences many genes/proteins that are also downstream targets of MYC and may be important to the mechanisms.

Strengths:

This multi-pronged approach found a previously unknown function for CAPSLs in endothelial cells and pointed at MYC pathways as high-quality candidates in the mechanism.

Weaknesses:

Two issues shape the overall impact for me. First, the unreported population frequency of the variants in the manuscript makes it unclear if CAPSL should be considered an interesting candidate possibly contributing to FEVR, or possibly a cause. Second, it is unclear if the identified variants act dominantly, as indicated in the pedigrees. The studies in mice utilized homozygotes for an endothelial cell-specific knockout, leaving uncertainty about what phenotypes might be observed if mice heterozygous for a ubiquitous knockout had instead been studied.

In my opinion, the following scientific issues are specific weaknesses that should be addressed:

(1) Please state in the manuscript the number of FEVR families that were studied by WES. Please also describe if the families had been selected for the absence of known mutations, and/or what percentage lack known pathogenic variants.

(2) A better clinical description of family 3104 would enhance the manuscript, especially the father. It is unclear what "manifested with FEVR symptoms, according to the medical records" means. Was the father diagnosed with FEVR? If the father has some iteration of a mild case, please describe it in more detail. If the lack of clinical images in the figure is indicative of a lack of medical documentation, please note this in the manuscript.

(3) The TGA stop codon can in some instances also influence splicing (PMID: 38012313). Please add a bioinformatic assessment of splicing prediction to the assays and report its output in the manuscript.

(4) More details regarding utilizing a "loxP-flanked allele of CAPSL" are needed. Is this an existing allele, if so, what is the allele and citation? If new (as suggested by S1), the newly generated CAPSL mutant mouse strain needs to be entered into the MGI database and assigned an official allele name - which should then be utilized in the manuscript and who generated the strain (presumably a core or company?) must be described.

(5) The statement in the methods "All mice used in the study were on a C57BL/6J genetic background," should be better defined. Was the new allele generated on a pure C57BL/6J genetic background, or bred to be some level of congenic? If congenic, to what generation? If unknown, please either test and report the homogeneity of the background, or consult with nomenclature experts (such as available through MGI) to adopt the appropriate F?+NX type designation. This also pertains to the *Pdgfb-iCreER* mice, which reference 43 describes as having been generated in an F2 population of C57BL/6 X CBA and did not designate the sub-strain of C57BL/6 mice. It is important because one of the explanations for missing heritability in FEVR may be a high level of dependence on genetic background. From the information in the current description, it is also not inherently obvious that the mice studied did not harbor confounding mutations such as *rd1* or *rd8*.

(6) In my opinion, more experimental detail is needed regarding Figures 2 and 3. How many fields, of how many retinas and mice were analyzed in Figure 2? How many mice were assessed in Figure 3?

(7) I suggest adding into the methods whether P-values were corrected for multiple tests.

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